

Characteristics of stars in NGC2301 using aperture photometry

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1 Abstract

The experiment uses two high-exposure images of the same section of sky using Monash University's Hutton-Westfold Observatory. One image is taken in the B-band and the other in the V-band. We conducted an analysis using the ds9 application which allowed us to determine the magnitudes of prominent stars in the image, which we could then use in a further mathematical analysis to determine their distance, surface temperature, and radius. We found that by comparing the data of our stars to that stored in the SIMBAD database, there were significant discrepancies in magnitude which resulted in incorrect distance and radius measurements. Some results were more accurate than others, and the inconsistent results were put mostly down to the presence of interstellar dust unaccounted for. We measured the distance to NGC2301 to be 511.47pc, 50.50% different than the SIMBAD literature of 872pc.

2 Introduction

Aperture photometry is an effective and widely used method to determine characteristics of distant stars. The key process is to take a small angular portion of the sky that covers a celestial object. By summing up all the "counts", the electrons hitting the instrument, we can determine a relative magnitude of the object (Mighell 1999). The purpose of this report is to demonstrate a simple application of aperture photometry as a means of determining the apparent magnitudes, distances, and radii of distant stars. Determining the distance using this method relies on the small angle approximation. Since the ratio between a star's diameter and the distance between Earth and the star is so large, the angular diameter will be equal to the angle across the star in the sky. This is an important principle used in this technique.

3 Method and Results

3.1 Apparent magnitudes in B- and V- bands

In Figure 1 we observe the same image of NGC2301 in two separate bands. In order to calculate the magnitude of the stars in each band we can use the measure of "counts" across an object. This refers to the amount of detected electrons over a particular area of an image. ds9 greatly simplifies this process using a function called Auto-Plot-Statistics. The first star we measured using this technique was located at the coordinates 6:51:46.18 +0:27:35. It is the uppermost bright star of the five bright stars at the centre of the image. The equations to find the magnitude in each band are given by $B = 23.058 - 2.5 \log_{10}(f_{\text{counts}})$ and $V = 23.397 - 2.5 \log_{10}(f_{\text{counts}})$, where f_{counts} is the measured "counts" over the object. For this chosen star,

$$B = 9.0232 \tag{1}$$

$$V = 8.1041 \tag{2}$$

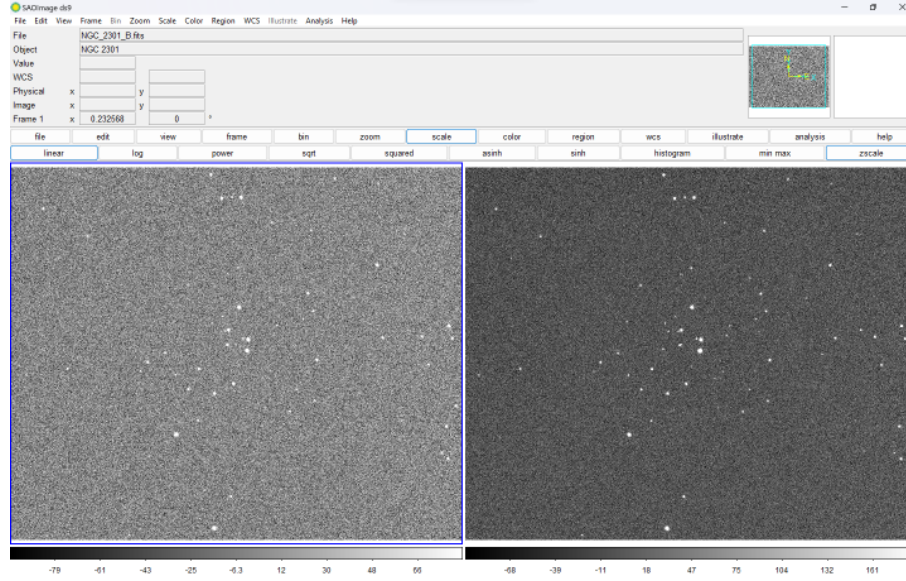


Figure 1: B-band image (left) and V-band image (right) of a section of stars in NGC2301 observed by the Hutton-Westfold Observatory loaded into ds9. Colours are scaled such that bright spots represent stars.

We can follow the same process for other stars in the image. Let's notice that at the centre of the image there are four other bright stars. These four stars will be analysed in addition to the first star. Table 1 lists the stars, their coordinates, and recorded magnitudes.

Star #	Coordinates	B-band magnitude	V-band magnitude
1	6:51:46.18 +0:27:35	9.0232	8.1041
2	6:51:44 +0:26:36.9	10.3852	10.3690
3	6:51:45 +0:28:57	9.0930	9.1164
4	6:51:44 +0:28:14.79	10.1880	10.1457
5	6:51:46 +0:27:56.5	9.0339	9.1018

Table 1: Brightest stars recorded from Figure 1 and their band magnitudes.

3.2 Distances to stars

To calculate the distance to these stars we will use the distance modulus formula $m - M = 5 \log_{10}(\frac{d}{10 \text{ pc}})$. This is motivated by the fact that the given data displays the magnitudes of the stars. In this formula, m refers to the apparent magnitude (calculated in the previous part), and M refers to the absolute magnitude of the star. Our aim is to find distance d in parsecs.

Our only remaining unknown is M . Figure 2 lists well-known star types as well as their 'color', or more correctly, their $B - V$ colour value. Since we have B and V , we can lookup the respective absolute magnitudes of our stars using their $B - V$ value.

Main-Sequence Stars (Luminosity Class V)									
Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	M_{bol}	BC	M_V	$U - B$	$B - V$
O5	42000	499000	13.4	60	-9.51	-4.40	-5.1	-1.19	-0.33
O6	39500	324000	12.2	37	-9.04	-3.93	-5.1	-1.17	-0.33
O7	37500	216000	11.0	—	-8.60	-3.68	-4.9	-1.15	-0.32
O8	35800	147000	10.0	23	-8.18	-3.54	-4.6	-1.14	-0.32
B0	30000	32500	6.7	17.5	-6.54	-3.16	-3.4	-1.08	-0.30
B1	25400	9950	5.2	—	-5.26	-2.70	-2.6	-0.95	-0.26
B2	20900	2920	4.1	—	-3.92	-2.35	-1.6	-0.84	-0.24
B3	18800	1580	3.8	7.6	-3.26	-1.94	-1.3	-0.71	-0.20
B5	15200	480	3.2	5.9	-1.96	-1.46	-0.5	-0.58	-0.17
B6	13700	272	2.9	—	-1.35	-1.21	-0.1	-0.50	-0.15
B7	12500	160	2.7	—	-0.77	-1.02	+0.3	-0.43	-0.13
B8	11400	96.7	2.5	3.8	-0.22	-0.80	+0.6	-0.34	-0.11
B9	10500	60.7	2.3	—	+0.28	-0.51	+0.8	-0.20	-0.07
A0	9800	39.4	2.2	2.9	+0.75	-0.30	+1.1	-0.02	-0.02
A1	9400	30.3	2.1	—	+1.04	-0.23	+1.3	+0.02	+0.01
A2	9020	23.6	2.0	—	+1.31	-0.20	+1.5	+0.05	+0.05
A5	8190	12.3	1.8	2.0	+2.02	-0.15	+2.2	+0.10	+0.15
A8	7600	7.13	1.5	—	+2.61	-0.10	+2.7	+0.09	+0.25
F0	7300	5.21	1.4	1.6	+2.95	-0.09	+3.0	+0.03	+0.30
F2	7050	3.89	1.3	—	+3.27	-0.11	+3.4	+0.00	+0.35
F5	6650	2.56	1.2	1.4	+3.72	-0.14	+3.9	-0.02	+0.44
F8	6250	1.68	1.1	—	+4.18	-0.16	+4.3	+0.02	+0.52

Main-Sequence Stars (Luminosity Class V)									
Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	M_{bol}	BC	M_V	$U - B$	$B - V$
G0	5940	1.25	1.06	1.05	+4.50	-0.18	+4.7	+0.06	+0.58
G2	5790	1.07	1.03	—	+4.66	-0.20	+4.9	+0.12	+0.63
Sun ^d	5777	1.00	1.00	1.00	+4.74	-0.08	+4.82	+0.195	+0.650
G8	5310	0.656	0.96	—	+5.20	-0.40	+5.6	+0.30	+0.74
K0	5150	0.552	0.93	0.79	+5.39	-0.31	+5.7	+0.45	+0.81
K1	4990	0.461	0.91	—	+5.58	-0.37	+6.0	+0.54	+0.86
K3	4690	0.318	0.86	—	+5.98	-0.50	+6.5	+0.80	+0.96
K4	4540	0.263	0.83	—	+6.19	-0.55	+6.7	—	+1.05
K5	4410	0.216	0.80	0.67	+6.40	-0.72	+7.1	+0.98	+1.15
K7	4150	0.145	0.74	—	+6.84	-1.01	+7.8	+1.21	+1.33
M0	3840	0.077	0.63	0.51	+7.52	-1.38	+8.9	+1.22	+1.40
M1	3660	0.050	0.56	—	+7.99	-1.62	+9.6	+1.21	+1.46
M2	3520	0.032	0.48	0.40	+8.47	-1.89	+10.4	+1.18	+1.49
M3	3400	0.020	0.41	—	+8.97	-2.15	+11.1	+1.16	+1.51
M4	3290	0.013	0.35	—	+9.49	-2.38	+11.9	+1.15	+1.54
M5	3170	0.0076	0.29	0.21	+10.1	-2.73	+12.8	+1.24	+1.64
M6	3030	0.0044	0.24	—	+10.6	-3.21	+13.8	+1.32	+1.73
M7	2860	0.0025	0.20	—	+11.3	-3.46	+14.7	+1.40	+1.80

Figure 2: An imaged copy of a table of main sequence stars with well-defined characteristics, including $B - V$ values and absolute magnitudes.

To determine the distance to the first star, let's observe that $B - V = 9.0232 - 8.1041 = +0.92$. The corresponding absolute magnitude M_V in the table is given by $+6.5$. Since the stars in the table are in discrete categories, and our magnitudes may not be perfectly accurate due to uncertainties explored further in this report, we can safely approximate the absolute magnitude by rounding to the nearest discrete value in the table (as opposed to using a method of increments to find the intermediary value).

Since our absolute magnitude is only given in the V-band, we also need to use the V-band apparent magnitude when using the distance modulus formula.

$$m - M = 5 \log_{10} \left(\frac{d}{10 \text{pc}} \right) \quad (3)$$

$$8.1041 - 6.5 = 5 \log_{10} \left(\frac{d}{10 \text{pc}} \right) \quad (4)$$

$$d_1 = 20.9324 \text{pc} \quad (5)$$

Table 2 lists the final distance measurements for each star

Star #	B-V Colour	Absolute Magnitude (V-band)	Distance (pc)
1	+0.92	+6.5	20.9324
2	+0.02	+1.3	651.328
3	-0.02	+1.1	401.125
4	+0.04	+1.5	535.969
5	-0.07	+0.8	457.467

Table 2: Distances of selected stars in parsecs using values from Figure 2.

We can immediately notice that the **first star is an outlier** in this data. The most reasonable conclusion to draw from this is that the star we measured is not a main sequence star. The data provided in Figure 2 assumes that our stars are main sequence stars. Since we know the star *is* a part of NGC 2301, it must not be a main sequence star. We will **exclude** this star from future calculations. Proceeding from this, the stars will be numbered 2-5 inclusive.

The **average distance** to the stars (excluding the first star) is

$$d_{\text{avg}} = 511.47 \text{pc}. \quad (6)$$

3.3 Calculating radius of stars

To motivate the following calculations, we will start by stipulating that the diameter of a star can be calculated using

$$\theta = \frac{\text{diameter (AU)}}{\text{distance (pc)}}, \quad (7)$$

where θ is the angular diameter. This comes from the small angle approximation by constructing the triangle between Earth and the star's diameter. To find

the angular diameter, we can again refer to a table of well defined values that lists stars based on their surface temperature and flux density at 0.01 and 0.1 milliarcsecond diameters. Figure 3 is a list of these well-defined values.

Temperature	V-band Flux Density (Jy) of 0.01 milliarcsecond diameter source	V-band Flux Density (Jy) of 0.1 milliarcsecond diameter source
26,000 K	0.257	25.7
24,000 K	0.226	22.6
22,000 K	0.195	19.5
20,000 K	0.165	16.5
18,000 K	0.135	13.5
16,000 K	0.107	10.7
14,000 K	0.0810	8.10
12,000 K	0.0564	5.64
10,000 K	0.0348	3.48
9,000 K	0.0254	2.54
8,000 K	0.0173	1.73
7,000 K	0.0107	1.07
6,000 K	0.0056	0.56

Figure 3: An imaged copy of a table of flux densities of blackbodies (in Jansky) as a function of temperature (Kelvin) and angular diameter (in milliarcseconds).

Our goal now is to determine the surface temperature of the stars, and use the table to lookup the corresponding V-band flux density of a blackbody with that surface temperature. We can also observe that, by using the table, a difference in 10x milliarcseconds corresponds to a 100x difference in V-band flux density. Hence, a difference of k times in flux density will be a difference of $\sqrt{k}$ times in angular diameter. Our recorded flux values in the V-band can be found using the formula $m = -2.5 \log_{10}(\frac{f_V}{f_{V,Vega}})$, where m is the apparent magnitude from earlier, and $f_{V,Vega}$ equals 3644 Jy.

To find the surface temperatures of our stars we can simply refer back to Figure 2 and record the corresponding temperature. This information is compiled in Table 3.

Star #	Surface Temperature (K)	Measured Flux f_V (Jy)
2	9400	0.2594
3	9800	0.8223
4	9020	0.3186
5	10500	0.8334

Table 3: Stars and their surface temperature (using Fig. 2) and calculated flux densities using the apparent magnitude in the V-band.

For Star 2, we recorded a surface temperature of 9400K. Observing Figure 3, we can see that temperature and V-band flux density changes linearly, therefore it is appropriate to use a method of increments to find the intermediary value between 9000K and 10000K that corresponds to 9400K. This is four-tenths past 9000K, so we can add four tenths of the difference between flux densities 9000K

and 10000K to the flux density at 9000K. We will follow the same process for our other three stars. $0.0253 + 0.4(0.0348 - 0.0254) = 0.0292$ Jy. Our measured V-band flux is $\frac{0.2594}{0.0292} = 8.88$ x greater, so our angular diameter will be $\sqrt{8.88 \cdot 0.01} = 0.0230$ milliarcseconds.

Using Equation (7) with this angular diameter and our distances from Part 3.2, we can determine the diameter of our stars in AU. In Table 4, we represent this as solar radii.

Star #	Angular Diameter (mas)	Diameter (AU)	Solar Radii
2	0.0230	0.01498	1.62
3	0.0500	0.02004	2.16
4	0.0353	0.01891	2.04
5	0.0455	0.02082	2.25

Table 4: Solar radius of stars using their V-band flux densities and the small angle approximation.

4 Discussion

It is appropriate to make a comparison between the recorded results of this method with those of a well-defined database, in order to determine shortcomings and inaccuracies of the method. We used the SIMBAD database which lists expected apparent magnitudes and radii of our stars. Before we begin, note that our measured distance to NGC2301 was 511.47pc. The accepted distance to NGC2301 is 857 pc (SIMBAD 2021). This is a 50.50% difference. Such a large discrepancy, as explored further in this discussion, should be placed down to the presence of interstellar dust affecting the recorded magnitudes of the stars.

Tables 5 and 6 include the results from the method for each star and compares these with the measurement obtained from SIMBAD.

Star #	B-band mag.	Expected B-band mag.	# uncertainties different	V-band mag.	Expected V-band mag.	# uncertainties different
1	9.0232	9.180	1.72%	8.1041	7.980	1.54%
2	10.3852	10.400 ± 0.03	1	10.3690	10.440 ± 0.04	2
3	9.0930	8.98	1.25%	9.1164	9.02	1.06%
4	10.1880	10.030 ± 0.03	6	10.1457	10.000 ± 0.04	4
5	9.0339	9.01 ± 0.02	1	9.1018	9.15 ± 0.02	3

Table 5: Band magnitudes of stars from Table 1 compared with expected results from SIMBAD. In the cases where the literature lists no uncertainty, percentage difference has been provided.

What we find is that our recorded measurements for the V- and B- band magnitudes of the stars are within very few uncertainties. The highest differences were the B- and V-band magnitudes of the fourth star.

Star #	Solar Radius	Expected Solar Radius	% difference
2	1.62	2.1	25.81
3	2.16	2.2	1.83
4	2.04	2.0	1.98
5	2.25	2.3	2.20

Table 6: Radius of stars from Table 4 compared with expected results from SIMBAD.

Interestingly, the second star is over 25% different from the literature, while the other stars are within a 1-3% difference. This is a significant discrepancy. Upon further analysis, due to the B-band wavelength range being inside the range for the visible light wavelengths, this will be heavily affected by the presence of interstellar dust between us (the observers) and the star. This means the "counts" that we initially recorded in the first part would be less than the actual amount. Such a discrepancy would affect the magnitude we record, the distance measurement to the star, and as we see, the recorded radius.

The use of aperture photometry to determine the radius and other characteristics of distant stars uses a method of small angle approximations and trigonometry. The small angle approximation is derived from the idea that for very, very small angles, the sin and tan values of these angles are approximately equal. This area of mathematics and astronomy is able to be highly automated by the use of algorithms and machine learning. An interesting study (Assafin 2023) done in December of 2023 explores the use of technology to quickly apply the principles of aperture photometry to vast amounts of heterogeneous data. Heterogeneous data refers to the idea that the data collected and presented by various international organisations will be very different in terms of how the data was collected, by which instruments, and for which purposes. The Package for the Reduction of Astronomical Images Automatically (PRAIA) indexes this data in order to make collaborations easier. One of the main tasks of the PRAIA photometry task is to perform aperture photometry over digitised FITS images (Assafin 2023). The system is suited for solar system bodies, distant celestial objects, and even stellar occultations. The system automatically does what we have done in this report. It takes count measurements over stellar objects and determines apparent variations in flux from these. This is a highly interesting extension of the principles applied in this report.

5 Conclusion

This report demonstrates the use of aperture photometry to determine characteristics of stars including their angular diameter, diameter, radius, distance, and magnitude.

We found an underestimated distance for stars in NGC 2301 to be at 511.47 pc, which was approximately 50% different from the SIMBAD literature. We also determined the solar radii of our stars to be approximately 2 solar radii.

Apart from our second star, our results were within a few percent of the literature. The discrepancies in our results would have been due to the presence of interstellar dust in the data recording process.

6 References

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